Customer Shopping Behaviour Analysis

Analysis of 3,900 purchases to uncover spending patterns, segments, product preferences, and subscription behavior to guide strategy.

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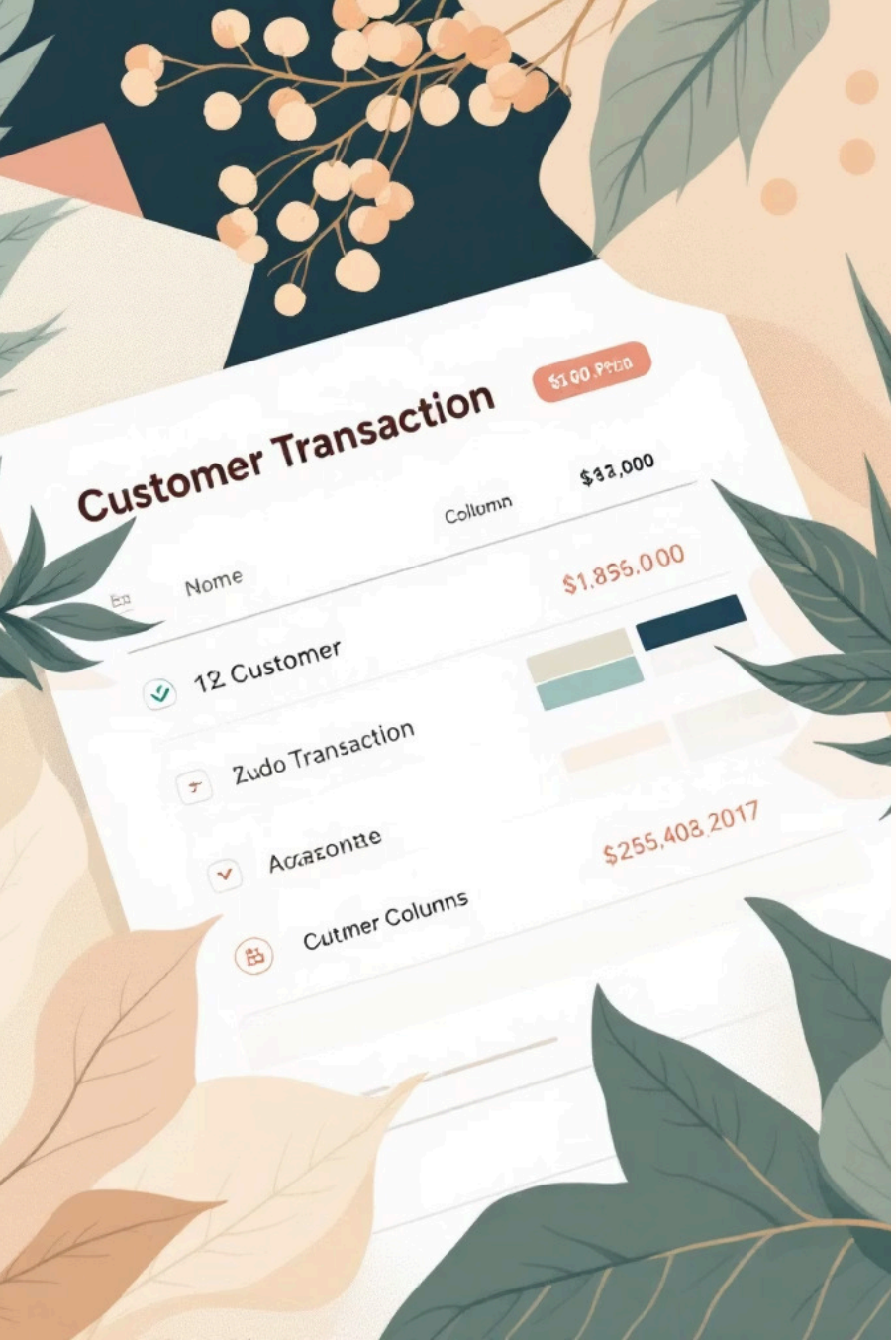

Dataset Snapshot

Rows & Columns

3,900 rows · 18 columns

Key Features

Demographics, purchase details, behaviour signals (discounts, frequency, ratings)

Missing Data

37 missing values in Review Rating

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Data Preparation (Python)

01

Load & Explore

Imported with pandas; used df.info() and describe().

02

Clean & Impute

Imputed Review Rating by product-category median; standardized column names to snake_case.

03

Feature Engineering

Created age groups and purchase_frequency_days.

04

DB Integration

Loaded cleaned DataFrame into PostgreSQL for SQL analysis.

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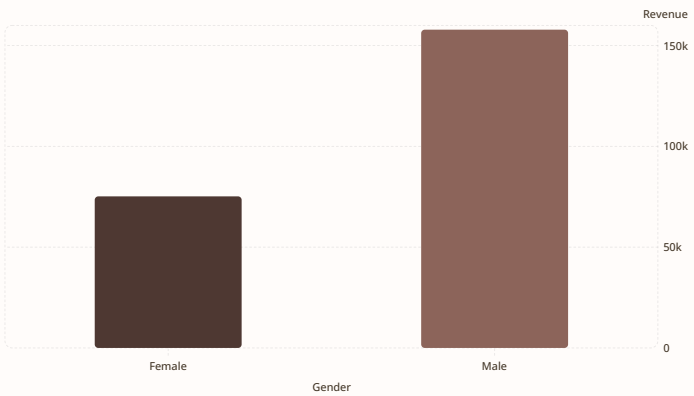
Revenue by Gender (SQL)

Male customers generated higher total revenue versus female customers. Use this to inform targeted promotions and product assortments.



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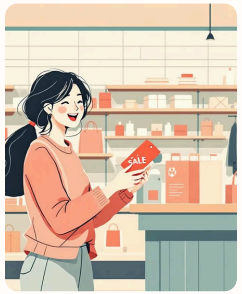
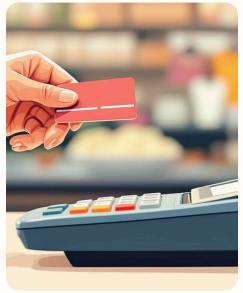
High-Spending Discount Users

Identified customers who used discounts but spent above average purchase amount — useful for refining discount and loyalty tactics.



Behavior Pattern

Discount users can still be high spenders; segmentation matters for margin impact.



Action

Target offers and rewards to high-spend discount users rather than blanket discounts.

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Customer Segmentation

3.1K

Customers

Total unique customers represented

3,116

Loyal

Majority of customers fall into Loyal segment

701

Returning

Repeat but not yet loyal

83

New

Recently acquired customers

Segment-specific strategies (loyalty, onboarding, reactivation) will maximize lifetime value.

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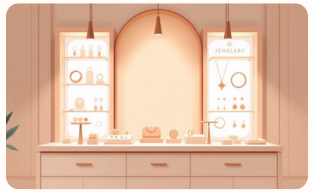
Top Products & Category Performance



Accessories

Top items: Jewelry (171), Sunglasses (161), Belt (161)

Clothing

Top items: Blouse (171), Pants (171), Shirt (169)

Footwear & Outerwear

Footwear: Sandals (160), Shoes (150), Sneakers (145). Outerwear: Jacket (163), Coat (161)

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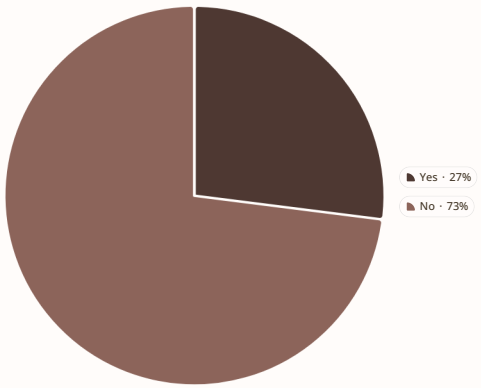
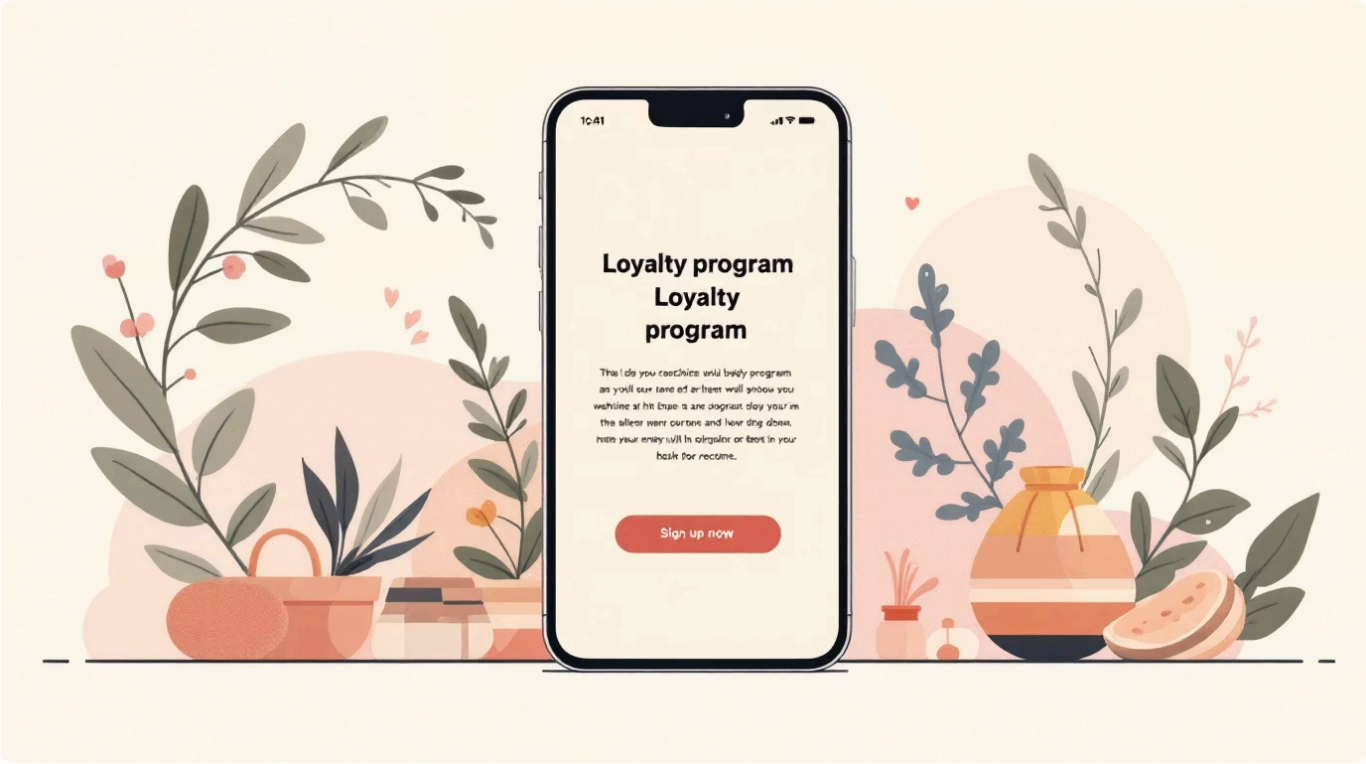
Repeat Buyers & Subscriptions



Repeat buyers (>5 purchases) are more likely to subscribe: Repeat buyers—Yes: 958; No: 2,518. Recommend targeted subscription incentives for repeat buyers.



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Power BI Dashboard Highlights

Key Metrics

3.9K customers · $59.76 avg purchase · 3.75 avg rating

Top Filters

Subscription, Gender, Category, Shipping Type

Category Snapshot

Revenue: Electronics 1000, Clothing 800, Books 500, Home Goods 200

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Business Recommendations

Boost Subscriptions

Promote exclusive subscriber benefits to increase conversion.

Loyalty Programs

Reward repeat buyers to grow the Loyal segment (3,116 customers).

Review Discount Policy

Balance short-term sales with margin control; target high-spend discount users.

Product Positioning & Targeting

Highlight best-sellers and target high-revenue age groups and express-shipping users.

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